

Exercise Prove **Isomorphism Theorem A**: If $\varphi : U \rightarrow V$ is a linear map, then

$$U / \ker \varphi \rightarrow \operatorname{im} \varphi, \quad \underbrace{u + \ker \varphi}_{=[u]} \mapsto \varphi(u)$$

is an isomorphism. Demonstrate *step-by-step* that the quotient map satisfies the following properties:

1. It is well-defined: $\varphi(u) = 0$ only if $[u] = [0]$;
2. It is injective: $\varphi(u) = 0$ is $[u] = [0]$;
3. It is surjective: every $y \in \operatorname{im} \varphi$ is the image of some $[u]$.

Example We shall apply **Isomorphism Theorem A** to establish **Isomorphism Theorem B**, which asserts that for any linear subspaces U and V :

$$\frac{U + V}{U} \simeq \frac{V}{U \cap V}.$$

Proof. We first define the map:

$$\ell : V \rightarrow \frac{U + V}{U}, \quad v \mapsto (v + U).$$

The map is surjective, since any $y \in \frac{U+V}{U}$ is of the form $u + v + U$, hence $y = (v + U)$ has a preimage v .

The kernel of ℓ consists of elements for which $\ell(v) = 0$, i.e., $v \in U$. Thus, $\ker \ell = U \cap V$. By **Isomorphism Theorem A**, we obtain:

$$\frac{V}{\ker \ell} = \frac{V}{U \cap V} \simeq \operatorname{im} \varphi = \frac{U + V}{U}.$$

Exercise Use **Isomorphism Theorem A** to prove **Isomorphism Theorem C**, which asserts that for inclusions of subspaces $U \subset V \subset W$, there exists an isomorphism:

$$\frac{W}{V} \simeq \frac{W/U}{V/U}.$$

答: 考虑映射

$$\Phi : W/U \rightarrow W/V, \quad (w + U) \mapsto (w + V).$$

1. 该映射是满的, 因为任意 $w + V$ 都有具有原像 $w + U$.

$$2. \ker \Phi = \{w + U \mid w + V = 0 + V\} = \{v + U \mid v \in V\} = \frac{V}{U}.$$

依照同构定理 A, $W/V = \operatorname{im} \Phi \simeq \frac{\operatorname{domain} \Phi}{\ker \Phi} = \frac{W/U}{V/U}.$

Exercise Use **Isomorphism Theorem A** to prove **Exercise 3.2** (见去年习题课讲义): Let $U_i \subset V_i$ be subspaces, prove the isomorphism

$$\frac{V_1 \times V_2}{U_1 \times U_2} \simeq \frac{V_1}{U_1} \times \frac{V_2}{U_2}.$$

答: 考虑映射

$$\Phi : V_1 \times V_2 \rightarrow \frac{V_1}{U_1} \times \frac{V_2}{U_2}, \quad (v_1, v_2) \mapsto (v_1 + U_1, v_2 + U_2).$$

最后例行公事: 判断满射, 找到 $\ker$, 使用同构定理 A.

Exercise Let $f : V \rightarrow V$ be a linear map. Use **Isomorphism Theorem A** to show that

$$\frac{\operatorname{im} f}{\operatorname{im} f \cap \ker f} = \operatorname{im} f \circ f = \frac{\operatorname{im} f + \ker f}{\ker f}.$$

答: 左 $\simeq$ 右是同构定理 B 的直接表述. 往证 中 $\simeq$ 右: 考虑

$$\Phi : \operatorname{im} f + \ker f \rightarrow \operatorname{im} f \circ f, \quad v \mapsto f(v).$$

最后例行公事: 判断满射, 找到 $\ker$, 使用同构定理 A.

Optional Exercise Explain the following identity:

$$\frac{\operatorname{im} f^p}{\operatorname{im} f^p \cap \ker f^q} \simeq \frac{\operatorname{im} f^q + \ker f^p}{\ker f^p}.$$

答: 上式来自 $\operatorname{im} f^{p+q}$ 的等价表述, 并使用了小技巧 $f^p \circ f^q = f^q \circ f^p$.

Then prove Fitting lemma:

1. (for linear endomorphisms) Let $f : \mathbb{F}^n \rightarrow \mathbb{F}^n$ be an endomorphism, then

$$\ker f^\infty = \ker f^n, \quad \operatorname{im} f^\infty = \operatorname{im} f^n,$$

and $\mathbb{F}^n = \ker f^\infty \oplus \operatorname{im} f^\infty$.

答: 考虑零空间的生长问题:

$$\begin{aligned} \ker f^k &= \ker f^{k+1} \\ \xRightarrow{\text{限制在子空间上}} \ker(f^k)|_{\ker f} &= (\ker f^{k+1})|_{\ker f} \\ \xLeftrightarrow{\text{基本逻辑}} \ker f^{k+1} &= \ker f^{k+2}. \end{aligned}$$

同样,

$$\begin{aligned} \operatorname{im} f^k &= \operatorname{im} f^{k+1} \\ \xRightarrow{\text{限制在子空间上}} \operatorname{im}(f^k)|_{\operatorname{im} f} &= (\operatorname{im} f^{k+1})|_{\operatorname{im} f} \\ \xLeftrightarrow{\text{基本逻辑}} \operatorname{im} f^{k+1} &= \operatorname{im} f^{k+2}. \end{aligned}$$

依照归纳, 对 $k \geq n$, 零空间与像空间不会增长. 最后证明 Fitting 引理

1. 由 $\frac{\operatorname{im} f^\infty}{\operatorname{im} f^\infty \cap \ker f^\infty} \simeq \operatorname{im} f^\infty$, 故 $\operatorname{im} f^\infty \cap \ker f^\infty$;
2. 由 $\frac{\operatorname{im} f^\infty + \ker f^\infty}{\ker f^\infty} \simeq \operatorname{im} f^\infty \simeq \frac{V}{\ker f^\infty}$, 故 $\operatorname{im} f^\infty + \ker f^\infty = V$.

2. (for matrices) Any $M \in \mathbb{F}^{n \times n}$ is similar to $\begin{pmatrix} D & O \\ O & N \end{pmatrix}$, where D is invertible and N is nilpotent.

答: 直和项 $\operatorname{im} f^\infty \oplus \ker f^\infty = V$ 都是 f -不变子空间, 这给出了分块矩阵. 再由

1. $f : \operatorname{im} f^\infty \rightarrow \operatorname{im} f^\infty$ 是单射, 从而是同构,
2. $f : \ker f^\infty \rightarrow \ker f^\infty$ 幂零,

这分别对应 D 可逆, 以及 N 幂零.

Exercise Let be $X \xrightarrow{f} Y \xrightarrow{g} Z$ linear maps with no additional assumptions. Show that

$$1. g^{-1}(g(f(X))) = \operatorname{im} f + \ker g;$$

答: ($\subset$ 方向) $y \in g^{-1}(g(f(X)))$ 是某一 $gf(x)$ 关于 g 的原像, 也就是 $g(f(x) - y) = 0$. 因此 $y \in \operatorname{im} f + \ker g$.

($\supset$ 方向) 任意 $f(x) + y \in \ker g$ 是 $gf(x)$ 关于 g 的原像.

$$2. f(f^{-1}g^{-1}(0)) = \operatorname{im} f \cap \ker g;$$

答: ($\subset$ 方向) 任意 $y \in f(f^{-1}g^{-1}(0))$ 既属于 $\ker g$, 又属于 $\operatorname{im} f$.

($\supset$ 方向) 任意 $f(x) \in \operatorname{im} f \cap \ker g$ 满足 $g(f(x)) = 0$, 从而 $f(x)$ 是 $\ker(g \circ f)$ 中某元素关于 f 的像.

3. Isomorphism Theorem B implies

$$\frac{g^*0}{f_*f^*g^*0} \simeq \frac{g^*g_*f_*X}{f_*X}.$$

答: 依照以上两问,

$$\frac{\ker g}{\operatorname{im} f \cap \ker g} = \frac{g^*0}{f_*f^*g^*0}, \quad \frac{g^*g_*f_*X}{f_*X} = \frac{\ker g + \operatorname{im} f}{\ker g}.$$

之后就靠同构定理 B.